ELECTRICAL / AUTOMATION / SYSTEMS

INFRA RESILIENCE

PROJECT 03

Root cause diagnosis when complex systems fail noisily.

A critical infrastructure fault diagnosis laboratory designed to distinguish initiating faults from cascades of alarms across sensors, drives, control logic and software layers.

PROPOSED RESEARCH / NOT YET VERIFIED

SYSTEM / RESEARCH ARCHITECTURE

1 STAGE
Process model

→

2 STAGE
Telemetry & logs

→

3 STAGE
Causal graph

→

4 STAGE
Engineer review

THE PROBLEM TO SOLVE

Can dependency aware diagnosis identify the origin of a multilayer industrial failure more accurately than threshold alarms alone?

TECHNICAL DISTINCTION

1. The interesting problem is not just alarm detection; it is separating cause from symptom.
2. Best aligned with your baggage / automation experience when scoped as a conveyor or material handling system.

WHAT THE EVIDENCE MUST SHOW

- ✓ Top 1 / top 3 initiating fault accuracy
- ✓ Detection latency and false alarm rate
- ✓ Performance under simultaneous faults and missing telemetry